

Team 29: Design and Implementation of an AVR-Class Microcontroller on an FPGA

Department: Electrical and
Computer Engineering
Capstone 2026

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ABSTRACT

Commercial microcontrollers are inexpensive and easy to program, but their internal real-time timing path is largely fixed by the vendor. This project develops a 32-bit fixed-cycle microcontroller in VHDL for the DE10-Lite FPGA. The design combines a multi-cycle finite-state machine, sixteen 32-bit registers, fixed 32-bit instructions, separate program and data memories, and a 32-bit memory-mapped bus. GPIO, a 32-bit timer, hardware PWM, and a prioritized interrupt controller are integrated as on-chip peripherals. Dedicated probe pins expose interrupt request, acknowledge, ISR entry, GPIO response, and PWM timing. The contribution is a transparent and measurable real-time control platform rather than a claim that FPGAs are universally superior to MCUs.

INTRODUCTION & OBJECTIVE

A soft-core processor is described in hardware description language, customized for an application, and synthesized into programmable logic [1]. This makes the complete control path visible from the register-transfer level to a physical output pin. The project asks whether a deliberately simple FPGA microcontroller can provide a shorter, bounded, and easier-to-audit request-to-response path than an Arduino Nano-class MCU while remaining understandable to students. Objective: Design, simulate, implement, and measure the FPGA MCU, then compare its digital I/O timing with an ATmega328P-class system using identical trigger and response definitions.

Implementation and Comparison Platform

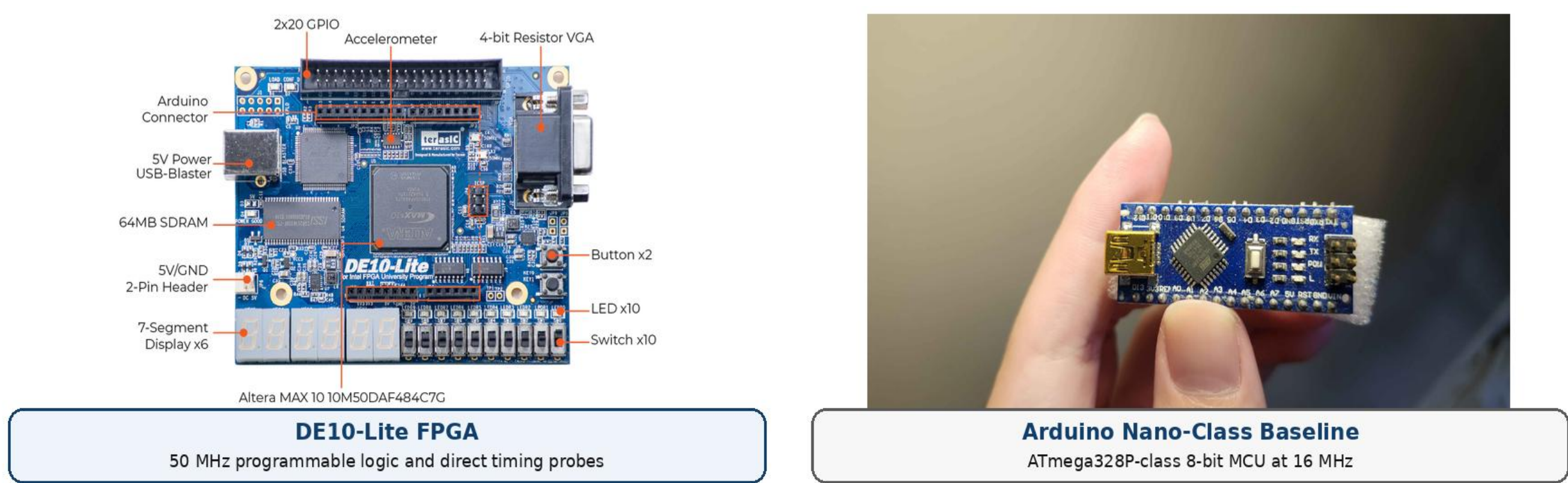


Figure 1: DE10-Lite implementation platform and Arduino Nano-class comparison baseline

DELIVERABLES

- 32-bit multi-cycle CPU with a fixed 32-bit instruction format.
- Sixteen 32-bit general-purpose registers and Z/N/C/V/I status flags.
- Separate 1024 × 32-bit program ROM and data RAM.
- Memory-mapped GPIO, timer, PWM, and prioritized interrupt controller.
- Self-checking VHDL testbenches, RTL Viewer diagrams, and timing constraints.
- Signal Tap and oscilloscope probes for interrupt latency and PWM verification.

DESIGN OVERVIEW

The CPU executes each instruction through explicit fetch, decode, execute, memory, and writeback states. A fixed-priority interrupt controller records pending sources, applies masks, selects a vector, and acknowledges service. GPIO, timer, PWM, interrupt, and debug registers share one fixed-cycle 32-bit memory-mapped bus. The PWM counter runs independently of firmware, reducing output jitter.

SYSTEM ARCHITECTURE

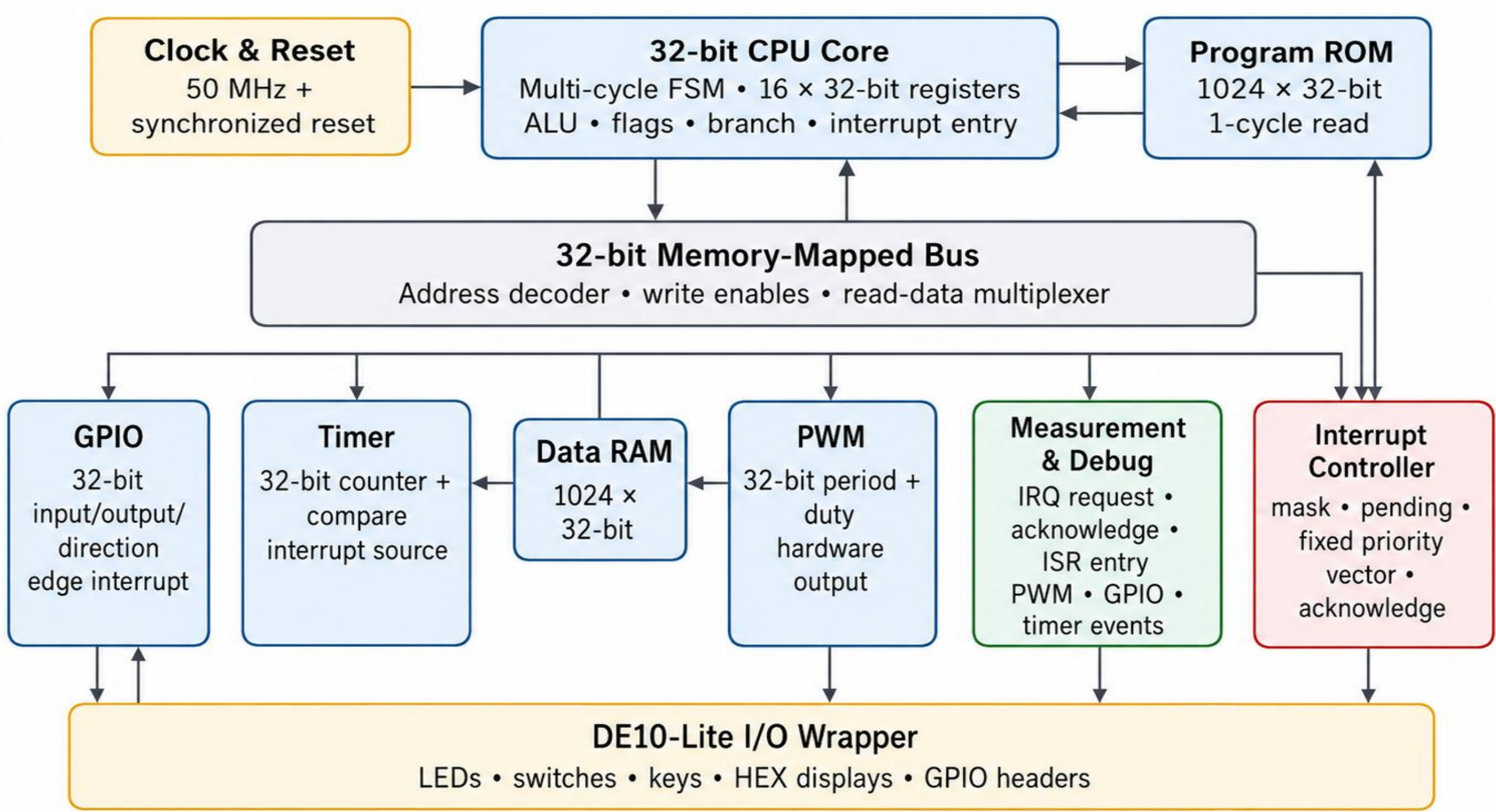


Figure 2: Top-level FPGA MCU architecture and memory-mapped peripheral structure.

METHODOLOGY

Development follows a simulation-first sequence. Clock/reset, ALU, register file, CPU, ROM, RAM, GPIO, timer, PWM, interrupts, and measurement logic are verified independently before full-system integration. The final design is synthesized in Quartus, constrained for 50 MHz, programmed through the USB-Blaster, observed in Signal Tap, and tested with the same oscilloscope event definition used for the Arduino.

Implementation & Verification Flow

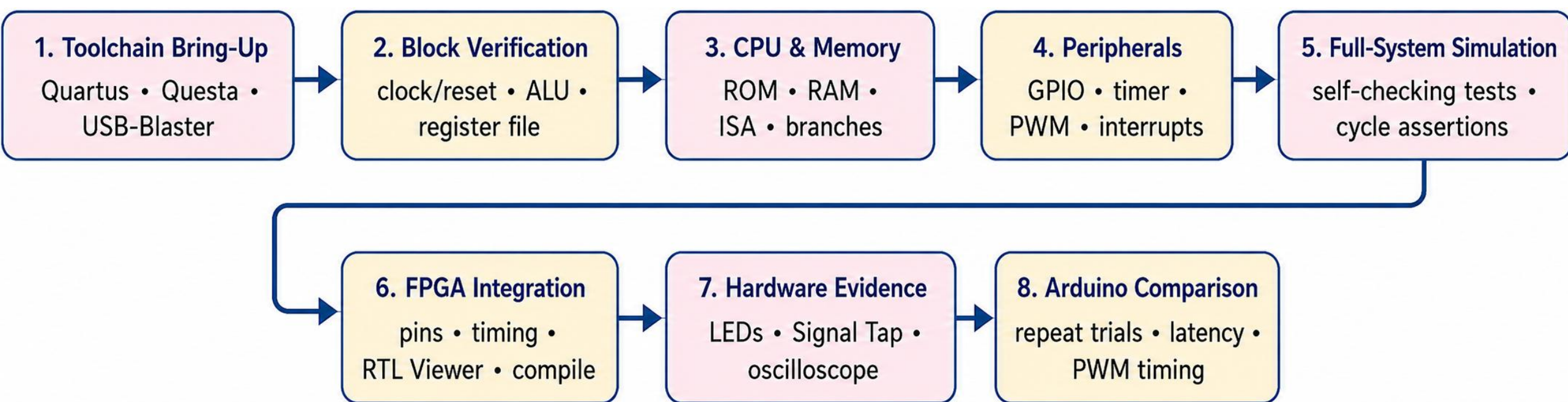


Figure 3: Block-by-block verification, FPGA integration, and hardware comparison workflow

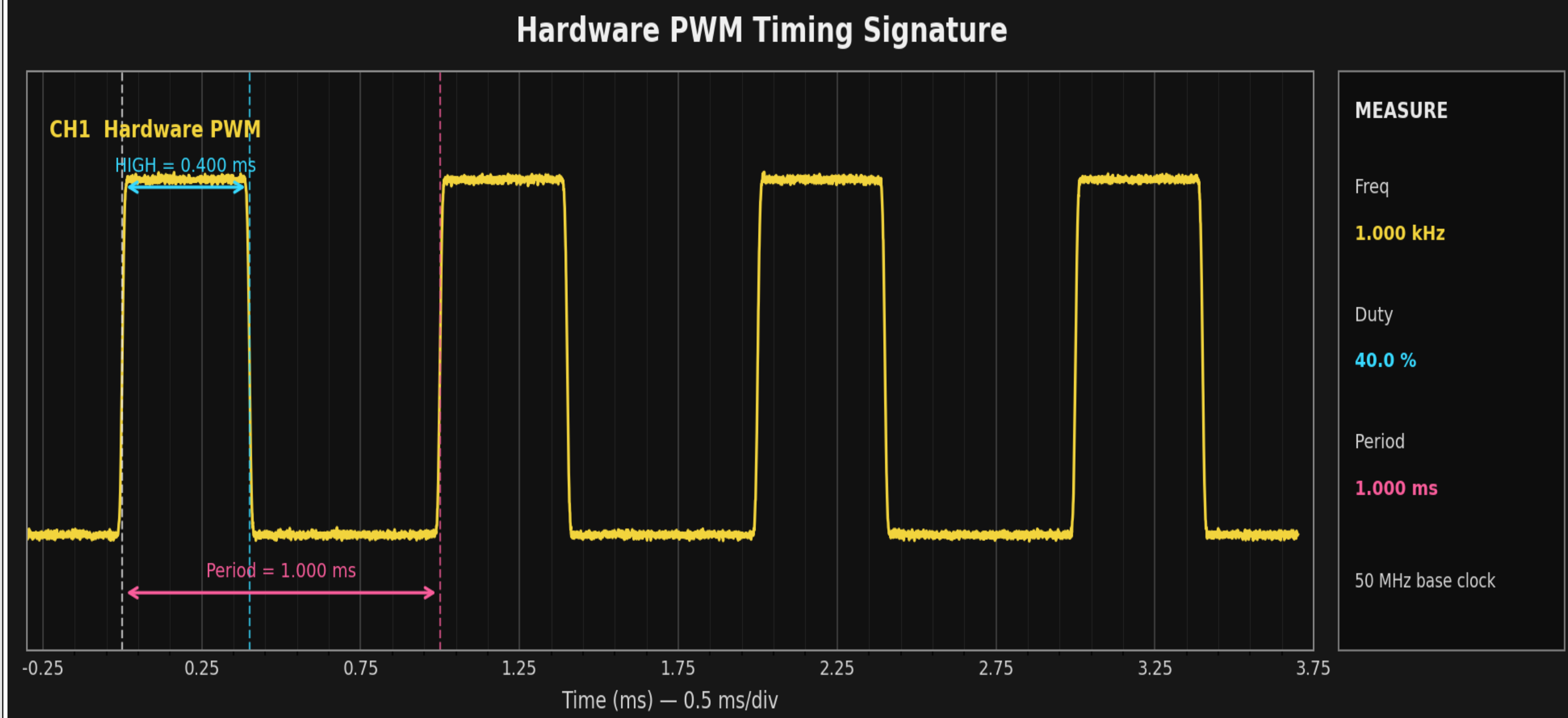
ARCHITECTURE

Feature	Design
Datapath / ISA	32-bit / fixed 32-bit
Register file	16 × 32-bit
CPU control	Multi-cycle FSM
Memories	1024 × 32-bit ROM and RAM
Peripherals	GPIO, timer, PWM, interrupts
Clock	50 MHz (20 ns/cycle)
Verification	Self-checking VHDL + Signal Tap + scope

RESULTS & TIMING COMPARISON

At 50 MHz, each clock cycle is 20 ns. The FPGA uses 3–5 cycles for normal instructions and up to 8 cycles for interrupt entry, giving delays of 60–160 ns. Including two synchronizer stages, the external-pin-to-ISR bound is 200 ns. The ATmega328P requires a 4-cycle interrupt response followed by a 3-cycle vector jump [2], totaling at least 437.5 ns at 16 MHz before the first handler instruction. Therefore, the FPGA target is approximately $2.2 \times$ faster.

INTERRUPT TIMING TRACE



TIMING SUMMARY

Metric	FPGA Design	Arduino Baseline
Clock	50 MHz	16 MHz
Instruction classes	60–100 ns	Architecture dependent
Core interrupt entry	≤160 ns	437.5 ns minimum + vector
External pin to ISR	≤200 ns target	≥437.5 ns; current instruction may add delay

CONCLUSION & FUTURE WORK

The project presents a 32-bit FPGA soft microcontroller with fixed-cycle execution, memory-mapped peripherals, hardware PWM, prioritized interrupts, and timing probes. Its clear timing behavior can be simulated, measured, and modified. Future work includes improved hardware testing, software tools, communication interfaces, expanded PWM and interrupt features, and comparison with a pipelined core.

ACKNOWLEDGEMENT & REFERENCES

We would like to thank Dr. Roberto Muscedere and our colleague Abdullah Mohammed Ali for their guidance and support.
[1] J. G. Tong, I. D. L. Anderson, and M. A. S. Khalid, “Soft-Core Processors for Embedded Systems,” ICM, 2006. [2] Microchip Technology, ATmega328P Data Sheet, Sec. 6.7.1. [3] Terasic Technologies, DE10-Lite User Manual.